3MB3 Final Project

Arms Race

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March 15, 2026

Overview

Scenario: Countries p and g are locked in an arms race where each is:

- Encouraged to increase military spending based on each other's spending
- Discouraged to increase spending based on their own their own spending

Model Overview:

- Δspending = x(other country's spending) - y(own spending)
- x and y measure the severity of encouragement or discouragement to spend

Research Question: What happens to the long-term spending of each country?

Arms Race Base Model

Model Equations

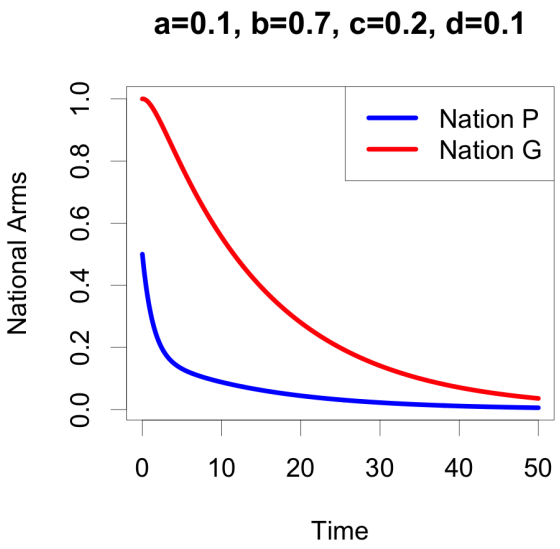
$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c: Mutual Fear Factor $(a, c > 0)$
- b, d: Expenditure Factor $(b, d > 0)$

Long Term Behavior

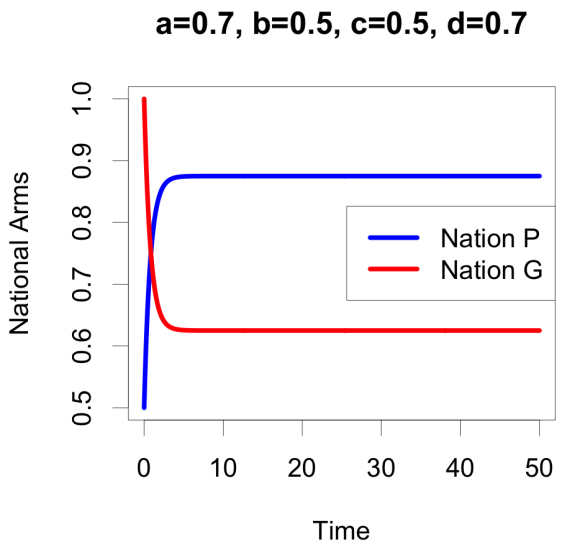
In the long run, we predict:

If: $bd > ac$



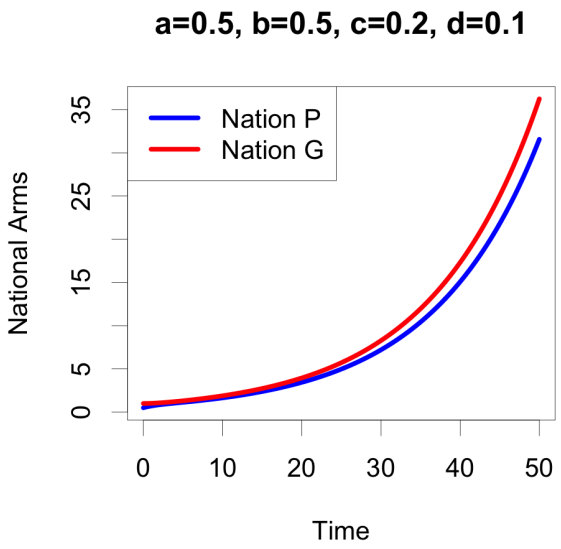
Complete mutual disarmament $(p^\infty, g^\infty) = (0, 0)$

If: $bd = ac$ **edge case**



Stabilized spending $(p^\infty, g^\infty) = (\frac{cp_0+bg_0}{a+c}, \frac{dp_0+ag_0}{a+c})$

If: $bd < ac$



Perpetual arms race $(p^\infty, g^\infty) = (\infty, \infty)$

Grievances and Goodwill

Suppose countries p and g have underlying goodwill/grievances that influence their spending
The model equations are therefore:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

- $(r, s) > 0$ for goodwill and $(r, s) < 0$ for grievance
- Provide further elaborating points

Long Term Behavior

- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods

Mutual Logistic Fear

Limitation of Current Model: Does not account for the limited resources each country has
Solution: Introduce logistic growth

$$\frac{\partial p}{\partial t} = a(1 - \frac{p(t)}{K_p})g(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = c(1 - \frac{g(t)}{K_g})p(t) - dg(t) + s$$

- Provide further elaborating points
- Provide further elaborating points

Long Term Behavior

- provide plots
- provide stability conditions
- provide convergences (if possible)

Thank you for listening

[0] Presentation theme by Ricardo B. Lourenco